Solutions to LLM + Robotic Oncology Trial Problems

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Abstract

Solving new large language model (LLM) and robotic based oncology clinical trial industry problems requires proper publication of protocols, standards, legislation, and funding applications prior to deployment. This paper provides solution sets derived from prior author works for upcoming Physical SI (super intelligence) oncology trials. The publications referenced here require whole implementation, as capabilities differ between state-of-the-art AI models and robots per problem type. Many AI manufacturers' models have been tested between 2024-2025 in order to discover novel and useful clinical trial applications. These developments include the creation of a patient matching machine learning pipeline, the combination of generated clinical trial meta-analyses to provide new AI insights for indications and cancer drugs, emerging clinical trial simulations, and accelerating the speed and quality of the academic paper creation process.

Ultimately, 2026 multi-prompt processes have been optimized to single master prompt systems featured in new trial protocols, drug applications, and funding requests - which feature autonomous AI sub-prompt generation and execution, professional machine readable diagrams, and iterations of full academic paper source code files using a repository-based LLM. Resulting single master prompt-based publications contained high contextual awareness, with a limited number of author corrections required for diagram positioning and paper formatting. Two comprehensive LLM and oncology clinical trial literature reviews conducted in 2025 and 2026 yielded a limited number applications, delayed implementations of new AI models - prompting the oncology trial solutions.

Keywords: Oncology clinical trials; Physical SI; Physical AI; Problem and solution; LLM

1 Introduction

Advancements in oncology clinical trials regarding LLMs and robotics have primarily been limited to patient trial matching applications and an extensive number of review articles [1, 2]. Additionally, 2025 and 2026 literature lacks end-to-end clinical trial phase applications using current AI models with greater competencies.

To that end, in June 2025 multiple pancreatic cancer related clinical trial meta-analyses were generated by ChatGPT from online sources. The meta-analyses then served as inputs to Gemini, which yielded a recommendation to perform research with cancer drug candidate daraxonrasib. A series of LLM-based trial, site, and sponsor simulations were conducted independently as daraxonrasib became FDA approved as Rasonque in August 2026. Overall survival (OS) from the author's daraxonrasib drug triplet quantitative systems pharmacology (QSP) trial simulation was 12.8 months vs. Revolution Medicines' 13.2 months Phase 3 daraxonrasib-only in-person trial (6.6 months OS chemotherapy control) [3, 4, 5].

Orderly accounts regarding project code [6], abstracts with publication visualizations, [7], LLM clinical trial application progressions [8], and a supporting presentation to this current work [9] aided solving foundational aspects of the new Physical SI industry.

Future LLM industry adaptions to works referenced in this publication are available to the pharmaceutical industry through large manuscript generation guidance [10], a principal investigator adoption guide [11], and several instructional documents [12]. LLM oncology clinical trial works throughout the year continued to build upon a January 2026 proposal [13], followed by LLM generated NIH clinical trial proposal [14, 15], LLM-automated Phase 1 applications [16], and a capitalization plan [17].

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2 Single Master Prompt with Sub-Prompts and Diagrams

Problem: Current multi-month publication and peer review processes negatively affect innovation and patient health. Kawchak K. utilized multi-prompt systems on the day-scale for academic paper publications in early 2026 where multiple prompt templates and author submissions were required.

Solution A: Single master prompt with LLM generated and executed sub-prompts mitigates extra steps on the hour-scale.
Solution B: Open-source machine readable diagrams converted to LaTeX in paper pdfs include: Mermaid, D2, GraphViz, PlantUML, Python Diagrams.
Solution C: Three full AI generated paper iterations per publication are provided as Draft, Full, and Final LaTeX source files.
Solution D: Repository and versioning updates were included within GitHub and Zenodo repositories [18, 19, 20, 21, 22, 23, 24, 25, 26] (last reference contains a follow-up user prompt).

3 Clinical Trial Site Simulations: Patients, Robots, Facility

Problem: Comprehensive and emerging oncology clinical trial site simulations are lacking due to slow industry 2026 LLM adoption.

Solution: Using detailed LLM prompts, quantitative simulations for patients, robots, and facilities on the hourly time scale in Python, or .txt and .md formats uploaded by AI to GitHub.

Papers: Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ [27].
- VVUQ Oncology Clinical Trial LLM Verification Automation Priority over Existing... [28].
- Threefold Humanoid 24/7 Adverse Event Oncology Trial Response Team: 4-Site Rotation [29].
- 2030: 60 Second Pancreatic Ductal Adenocarcinoma Robotic Whipple Procedure... [30].
- 2030: 60 Second Glioblastoma AI Robotic Surgery [31].
- Accelerated Patient Prediction in Physical AI Oncology Clinical Trials: 4 Extensive Simulations [32].
- A Cancer Patient's Journey Through a Regulated Physical AI Oncology Trial [33].

Demo: Four Station Oncology Clinical Trial Robot Competition Interactive Demo [34].

4 Clinical Trial Sponsor Simulations: AI-Human Exchange

Problem: Automating the workforce with agents is challenging due to complexities in replacing specific human roles/tasks/goals.

Solution: Used Claude Code's agents to simulate execute human pharmaceutical clinical trial tasks.

Papers: Accelerated Patient Prediction in Physical AI Oncology Clinical Trials: 4 Extensive... [32].

- Fully Automated Sponsor: Physical AI Oncology Clinical Trial Platform [35].

5 New 2026 H. R. Physical AI Oncology Trial Legislation

Problem: Previous forms of AI have not been able to analyze legislative and medical text inputs simultaneously while generating new bills on the hour time scale.

Solution: Used Claude Code across five individual bill versions to quickly and thoroughly address verifications, clinicians, legislators, and patients - all published within 2 weeks online: 5-30-26 to 6-10-26. Three additional works promoted the H. R. 9510 Bill from each of the different perspectives.

Papers: H. R. 9510 (Bill v5.0) 2026 [36].
- H. R. 9510 (Bill v4.0) 2026 [37].
- H. R. 9510 (Bill v3.0) 2026 [38].
- Verification Before Generation Act of 2026 [39].
- VVUQ Physical AI Oncology Trial Bill [40].
- Earning the Congress's Vote: A New Oncology Trial Framework for Enacting H. R... [41].
- Earning the Clinician's Trust: New Framework for Verified Physical AI Oncology Trials [42].
- From H. R. 9510 to Federal Law: A Narrative Case for Verified Physical AI Oncology Trials [43].

6 FDA CFR, ICH Documents Adapted to Physical AI

Problem: Full length FDA clinical trial guidance and regulations would likely require experts across multiple disciplines to incorporate LLMs and robotics while maintaining coherence with the FDA's online regulatory documents.

Solution: Used AI with author inputs of separate ICH guidance and CFR regulations to adapt new Physical AI oncology trial versions.

Papers: Adaption: 21 CFR Part 312, End-to-End Physical AI Oncology Clinical Trial... [44].
- Adaption: 21 CFR Part 50, End-to-End Physical AI Oncology Clinical Trial Unification [45].
- Adaption: ICH Harmonised Guideline, End-to-End Physical AI Oncology Clinical Trial... [46].

7 New Investigational Drug Application (IND)

Problem: IND applications are long and complex documents that take time and numerous considerations for large organizations to produce prior to commencing Phase I clinical trials.

Solution: Utilized a repository-based LLM to iterate through end-to-end machine readable diagrams and full paper drafts on the day-scale for a faster and illustrated work compared to human INDs.

Paper: Investigational New Drug Application - Daraxonrasib, Phase 1, LLM-Directed Robotic Whipple in KRAS-Mutated PDAC [47].

8 Pancreatic Cancer Phase 1 Clinical Trial Protocol

Problem: Trial protocols based on drug candidate progression are critical and challenging for setting up technical aspects of trials.

Solution: Utilized a repository-based LLM to finish an initial Phase 1 draft. Future protocol modifications can be made effortlessly when changes and additional data becomes available.

Paper: On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma [48].

9 Pancreatic Cancer Phase 2 Clinical Trial Protocol

Problem: Future trial protocols are difficult due to initial Phase 1 trials with real patients that have not yet finished.

Solution: Trial protocols based farther into the future can be finished within hours. As patient results are obtained from prior phases, the proposed protocol can be updated with low time commitments, and may also be used to advance funding purposes.

Paper: A Phase 2, Daraxonrasib + LLM Guided Robotic PDAC Whipple Procedure, Clinical... [49].

10 Physical AI Guidance, Guides for Next Oncology Trials

Problem: Few oncology trial developers are expected to have mastered all of the following AI proficiencies: a) conversational AI interfaces, b) prompt engineering, c) skills to work with disparate input file types, d) understanding of repository-based LLM limitations, and e) specifying optimal repository output formats for future editing and use as inputs in new projects.

Solution: Several guides and guidance were either AI generated or written by the author to prepare oncology field developers for clinical trial grade conversational AI applications.

Papers: Phase 1 Pancreatic Cancer Trial: Efficient LLM Document Guidance [10].
- Oncology Trial PI LLM Adoption Guide [11].

11 Site Move-In Packages: San Diego, San Francisco

Problem: Relevant regulatory documents are needed to initiate a clinical trial cities in high demand by patients.

Solution: Utilized a repository-based LLM to deliver 11-15 applicable clinical trial site move-in documents across two separate locations.

Papers: La Jolla Move-In: Pancreatic Oncology Clinical Trial Site Complete Documentation... [50].
- Physical AI Oncology Clinical Trial Site Complete Documentation Package [51].

12 Multiple NIH/NSF/DOE Funding Applications

Problem: Government funding applications require organization-specific knowledge not immediately apparent to innovators of new industries.

Solution: Claude Code was utilized to generate two versions of a NIH Director's Pioneer Award application pdf. Many comprehensive funding email applications, and numerous relevant follow-up email messages have been delivered - many containing additional AI generated papers/document attachments.

Papers: Clinical Trial Funding Application v2.0, RFA-RM-27-001, Kawchak K. [15].
- Clinical Trial Funding Application, RFA-RM-27-001, Kawchak K. [14].
- 10 Funding Applications: A Phase 1, First-in Human, PDAC Clinical Trial Protocol of a LLM-Directed Robotic Whipple with Daraxonrasib (RMC-6236) [16].

13 Patient Robot Advocacy to Protect Human Rights

Problem: Patients considering upcoming Physical SI/AI oncology trials will likely have many concerns regarding the safety and function of state-of-the art AI and robotics.

Solution: Created patient advocacy documents to assist patients' technology concerns in regards to

their upcoming cancer treatments.

Papers: Patient Robot Advocacy: A Phase 1, First-in-Human, PDAC Clinical Trial Protocol of a LLM-Directed Robotic Whipple with Daraxonrasib (RMC-6236) [52].
- Patient Priority of Proposed U.S. Bills for Physical AI Oncology Clinical Trials [53].
- National Platform for Physical AI Oncology Trials [54].
- A Cancer Patient's Journey Through a Regulated Physical AI Oncology Trial [33].

14 Verification, Validation, Uncertainty Quantification (VVUQ)

Problem: LLM workflows have not fully adapted effective and standardized VVUQ policies for emerging oncology clinical trials.

Solution: Used prior VVUQ oncology trial frameworks to adapt several new Physical AI bill versions, and emerging simulation platforms.

Papers: Verification Before Generation Act of 2026 [39].
- VVUQ Physical AI Oncology Trial Bill [40].
- Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ [27].
- VVUQ Oncology Clinical Trial LLM Verification Automation Priority over Existing Generated... [28].

15 2030 Pancreatic, Glioblastoma 60s Robotic/AI Surgeries

Problem: Current pancreatic cancer and glioblastoma surgeries are limited by the speed of robot and human intelligence and physical movements.

Solution: Author simulations detailed how a quickly surgical robot needs to operate alongside an onsite repository-based LLM in order to complete respective pancreatic cancer and glioblastoma surgeries within 60 seconds.

Papers: 2030: 60 Second Pancreatic Ductal Adenocarcinoma Robotic Whipple Procedure & Daraxonrasib Simulation [30].
- 2030: 60 Second Glioblastoma AI Robotic Surgery [31].

16 National Platform for Physical AI Oncology Trials

Problem: A new conversational AI and robotics platform for emerging oncology trials has not been fully elucidated.

Solution: A 186-page document utilized the author's Physical AI works and diagrams to produce a stepwise set of guidelines to the new industry. The publication is aimed at educating and inspiring existing oncology trial leaders.

Paper: National Platform for Physical AI Oncology Trials [54].

17 National MCP Servers for Upcoming Physical AI Trials

Problem: MCP servers have been used by AI manufacturers across technology companies, but no known foundation regarding LLM and robotics infrastructure has been formalized by the oncology trial industry. Current clinical trial point to point system compatibility will be improved when scaled with MCP servers.

Solution: Generated a five-server MCP deployment plan with an interactive GitHub code demonstration to modernize the current oncology trial system with greater cross-trial and cross-robot interoperability.

Papers: National MCP Servers for Physical AI Oncology Clinical Trial Systems [55].
- TrialMCP: MCP Servers for Physical AI Oncology Clinical Trial Systems [56].

Demo: National MCP Servers for Physical AI Oncology Clinical Trial Systems Interactive Demo [57].

18 Federated Learning Adaption to Robotics/AI

Problem: Federated learning across multiple sites has been effective at reducing patient data biases associated with one location, but has not been utilized with performant conversational AI models run on clinical trial robots.

Solution: Generated a 235 Python module framework with five critical infrastructure pillars that represents a robust and likely first physical AI oncology trial federated learning platform.

Paper: Federated Learning Physical AI Oncology Trials Unification [58].

19 Unification Standard Level (USL), Physical AI Standard Level (PSL)

Problem: Physical AI software, robotics, and interoperability standards have not been formalized for emerging oncology clinical trials. Additionally, robot site level compliance has not been defined for cancer related trials.

Solution A: Unification Standard Level (USL) governs four metrics regarding a robot's ability to switch simulation frameworks (Isaac Lab, MuJoCo, etc.), integrate with generative AI and agents, cross-robot progress sharing, and multi-site clinical trial cross collaboration.
Solution B: Physical AI Standards Level (PSL) evaluates site level compliance for each robot's legislative authorization, regulatory compliance, and operational readiness.

Papers: Unification Standard Level for Physical AI Oncology Trials [59].
- Physical AI Oncology Clinical Trial Site Complete Documentation Package [51].
- National Platform for Physical AI Oncology Trials [54].

20 Code Generation Competition for Adverse Events

Problem: Large scale LLM code generation comparisons across 9 open and proprietary manufacturers for patient adverse events (AEs) is an important area to test AI performance.

Solution: Eight open source and eight proprietary LLM generated code based on an AE problem were evaluated head to head, with a top proprietary model by OpenAI defeating a top open source model by OpenAI.

Paper: Code Generation Competition: 16 Proprietary vs. Open-Source LLMs & Iterative Learning Based on FDA Adverse Event Reporting System [60].

21 AI Peer Review for Clinical Trial Patient Matching

Problem: LLM peer-reviews of published papers have not been standardized for oncology clinical trial applications.

Solution: An AI peer-review system quality, speed, and cost benefits for complex and voluminous data was established for adaption in all future AI generated papers.

Papers: AI Peer Review Acceleration of LLM-Generated Glioblastoma Clinical Trial Patient Matching ML, FDA/ICH/ISO, and FastAPI [61].

- Federated Learning Physical AI Oncology Trials Unification [58].

22 LLM Task Substitutions for Clinical Trial Workflows

Problem: Oncology trial leaders and author implementations of LLM workflows have not been completed for emerging oncology clinical trials.

Solution: Many oncology trial field and author publications of LLM applications were organized as replacements or upgrades to current trial tasks. Time and cost predictions by Claude projected a multi-million cost savings when utilizing LLM incorporations for a glioblastoma clinical trial.

Paper: End-to-End Oncology Clinical Trial LLM Efficiency For Industry Adoption with FDA... [2].

23 Digital Twin Pancreatic Cancer Simulation

Problem: Digital twin implementations for pancreatic cancer patients represent a promising future application, however open source AI generated Python notebook code for a pancreatic cancer trial based on daraxonrasib was not yet been openly published.

Solution: LLM models were used to generate a multi-drug, bidirectional digital twin pancreatic cancer clinical trial simulation.

Paper: Accelerating FDA Compliance and Cost Efficiency of in silico Clinical Trials via AI Digital Twin Pancreatic Cancer Simulation [62].

24 QSP Pancreatic Cancer AI Clinical Trial Simulation

Problem: Rigorous QSP (Quantitative Systems Pharmacology) simulations are common to prove drug utility prior to in-person clinical trials, however a LLM code generation application at scale could not be identified.

Solution: LLMs were capable of generating the full complex QSP multi-drug clinical trial simulation.

Paper: QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation From Protocol to Prediction: Code, VVUQ, and Playbook [5].

25 In Silico Phase Pancreatic Cancer Clinical Trial

Problem: Can ChatGPT generate results for a 100,000 patient code-free pancreatic cancer clinical trial simulation?

Solution: The LLM produced results for the trial, however some variations in reproducible existed run across triplicate conversations, as expected due to non-deterministic use with the AI model.

Paper: ChatGPT 100,000 Patient 24-Month In Silico Phase III 5-Arm Pancreatic Cancer Clinical Trial Triplicate [63].

26 Pancreatic Cancer Digital Twin Clinical Trial Proposals

Problem: Clinical trial proposals require much research and time when performed by humans.

Solution: Research paper first employed 40 ChatGPT meta-analyses that were inputted into Gemini which recommended future FDA approved daraxonrasib (Rasonque) for ChemicalQDevice to receive funding. Several LLMs judged each of their own outputs to determine top pancreatic cancer digital

twin proposals. ChatGPT, followed by Claude produced the highest overall quality proposals.

Paper: End-to-End Pancreatic Ductal Adenocarcinoma Digital Twin Clinical Trial Proposals [64].

27 Glioblastoma Clinical Trial Meta-Analyses at Scale

Problem: Are LLM-derived meta-analyses consistently generated across a single research project?

Solution: Multiple contextually aware LLM meta-analyses were produced in this publication.

Paper: 10 Year Glioblastoma Clinical Trial Meta-Analyses by Autonomous AI at Scale. Survival, HR, AE, and RoB scored in AI Reports and Charts, including Verifications [65].

28 AI Towards the Cure of Lung Cancer

Problem: A cure to lung cancer is challenging due to reconciling research data originating from disparate groups and journals. Can a technology be used that reconciles these differences while offering novel insights to cure lung adenocarcinoma?

Solution: LLMs successfully synthesized information from multiple lung adenocarcinoma sources, and offered understandings to cure this cancer type, based on inputs provided by the author.

Paper: AI revolution toward the cure of lung adenocarcinoma [66].

29 Cancer Drug Cost & Cancer Industry Cost

Problem: Many factors affect the cost of cancer drugs and industry economics associated with clinical trials. A technology is needed that reads multiple papers concurrently, and provides coherent solutions.

Solution: LLMs were utilized to provide cited summaries of articles, as well as novel financial insights.

Papers: Autonomous LLM Agent and scalable Reasoning LLM for generating cancer drug industry cost solutions [67].
- Cost containment of global monoclonal antibody drugs and cancer clinical trials via LLM... [68].

30 Clinical Decision Support: Bevacizumab Cancer Trials

Problem: Can an early 2025 LLM be used to summarize 100 separate research articles identified by the author, with a separate LLM drawing new insights from the collective summaries regarding how Bevacizumab was used across multiple cancer types?

Solution: Claude Sonnet was used to summarize each paper independently; and the ChatGPT o1 reasoning model provided practical cancer drug insights considering each of the 100 Claude summaries.

Paper: Clinical decision support based on Bevacizumab cancer trials and pushing the limitations of advanced LLMs [69].

31 Cancer vs. Conversational Artificial Intelligence

Problem: The cancer field is unaware of advanced LLM context and diagram functionalities.

Solution A: Provided LLM-generated Python tumor immune microenvironment knowledge graph image based on relevant cancer related literature.
Solution B: Included LaTeX-based tumor cell signaling pathway image based on cancer mechanism information.

Solution C: An additional combined tumor immune microenvironment signaling influences pathway image was also included, which was based on providing the LLM with updated cancer mechanism text.

Paper: Cancer vs. Conversational Artificial Intelligence [70].

32 Paclitaxel Biosynthesis AI Breakthrough

Problem: The full cancer drug paclitaxel biosynthesis pathway was still largely under development at the time of publication in 2024.

Solution A: Several recent leading paclitaxel papers were provided to the LLM for analysis and generation of new biosynthesis and retrobiosynthesis pathway recommendations.
Solution B: In 2025, the field made additional progress towards a complete paclitaxel biosynthesis pathway [71].

Paper: Paclitaxel Biosynthesis AI Breakthrough [72].

33 LMM Chemical Research with Document Retrieval

Problem: Could LLMs be used to analyze chemical and biochemical literature with high fidelity?

Solution: LLMs exhibited accurate capabilities in quickly evaluating and providing additional insights based on author provided papers.

Paper: LMM Chemical Research with Document Retrieval [73].

34 Conclusions

In summary, large language models (LLMs) have been utilized to acquire clinical trial literature understanding in order to provide new oncology and financial insights. Python notebook clinical trial simulations and a patient matching machine learning pipeline have also been accomplished using OpenAI [74], Anthropic [75], and Gemini [76] AI models. Comprehensive single master prompts have been adapted from multi-prompt approaches to solve increasingly larger trial problems throughout 2026 studies. These works have been made increasingly practical through repository-based [77] LLMs enabling hour and day scale publications to address delays associated with clinical trial move-in packages, new legislation, and FDA guideline adaptions.

Finally, an automated solution to new funding applications has reduced months of comprehensive email requests - which also specify which author papers to attach, and often include additional recipient-specific pdf documents [18, 19]. As a result, the problem regarding lack of email responses has been addressed with increased numbers of productive funding organization responses when compared to author written communications.

35 Appendix

Note: USL and PSL Standards are derived from established technology readiness level (TRL) methodologies implemented by the U.S. Government:
1. NASA/DOD Technology Readiness Levels (TRL) - Mankins, J.C. (2004). Technology Readiness Assessments: A Retrospective. White Paper, NASA. Original 9-level TRL scale for evaluating technology maturity from basic principles (TRL 1) to flight-proven systems (TRL 9). USL adapts this graduated-maturity concept to robot unification readiness [78].
2. ML Technology Readiness Levels (MLTRL) - Lavin, A., et al. (2021). Technology Readiness Levels for Machine Learning Systems. GitHub: ai-infrastructure-alliance/mltrl. Extends TRL to ML systems with levels 1-9 covering data readiness, model development, deployment, and monitoring. USL incorporates MLTRL's recognition that AI system readiness requires evaluating software, data, and

integration dimensions beyond hardware alone [79].
3. TRL for Complex System Integration - Tomaschek, K., Olechowski, A., Eppinger, S., & Joglekar, N. (2015). A Survey of Technology Readiness Level Users. Proceedings of PICMET 2015. DOI: 10.1109/PICMET.2015.7273196. Identifies challenges in applying TRL to integrated multi-technology systems, directly relevant to evaluating robot systems that span simulation, AI, and clinical deployment [80].
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37 Ethical Disclosures

This is an independent collection of oncology clinical trial solutions. No real patients were enrolled, no protected health information was used, and no IRB approval was sought or obtained in auwthor references. This solution set is not an active IND or IDE, is not an approved protocol, and is not medical or regulatory advice. It is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor.

38 Cite This Article

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